

Progress Report: Five Grokking Predictors on the 5-Seed Nanda-Unified Run

Thesis: *A Unified Benchmark of Grokking Predictors in Neural Networks*

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1. Summary

- The 5-seed training run of the Nanda-Unified protocol (modular addition, $p = 113$, 40000 epochs) is complete. All 5 seeds grokked; grok epochs range from 6921 to 23747.
- Five predictors are now computed on all 5 seeds and scored against the grok epoch: **L2 Norm, Dropout, Spectral, AGE and HTSR Alpha** (Predictors 1 to 5 in the evaluation order).
- **Result: none of the five predictors works as implemented.** Each one either fires at almost the same epoch in every seed regardless of when grokking happens (L2 Norm), or reaches its extreme value only after the model has already grokked (Dropout, Spectral, AGE, HTSR).
- The underlying signals are not uninformative: they all change sharply around the grok epoch. It is the published "extreme value" reading that arrives late.
- Scope decision: the study benchmarks established predictors with their published methods. No new detection rule or tuned threshold is introduced. This is recorded as a limitation and as future work.
- Next in the evaluation order: Correlation Traps, then Weight-PCA, Higher-MI and Commutator Defect.

2. Experimental setup

One shared protocol is used for every predictor: 1-layer transformer, 4 attention heads, $d_{\text{model}} = 128$, no LayerNorm, small initialisation $N(0, 0.8/\sqrt{d_{\text{model}}})$, AdamW (learning rate 0.001, betas 0.9 and 0.98, weight decay 1.0), full-batch training, 30/70 train/test split, Apple Silicon (MPS). Grok epoch is defined as the first epoch with test accuracy above 0.9. Each seed took about 2.1 hours to train; all 5 seeds reach train and test accuracy of 1.0 and none shows a sustained post-grok limit cycle.

Seed	0	1	2	3	4	Mean (std)
Grok epoch (test acc > 0.9)	12987	6921	9510	11019	23747	12837 (5804)

The grok epoch varies by a factor of 3.4 across seeds, with seed 4 a clear late outlier. This spread is what makes the test meaningful: a genuine predictor must follow it.

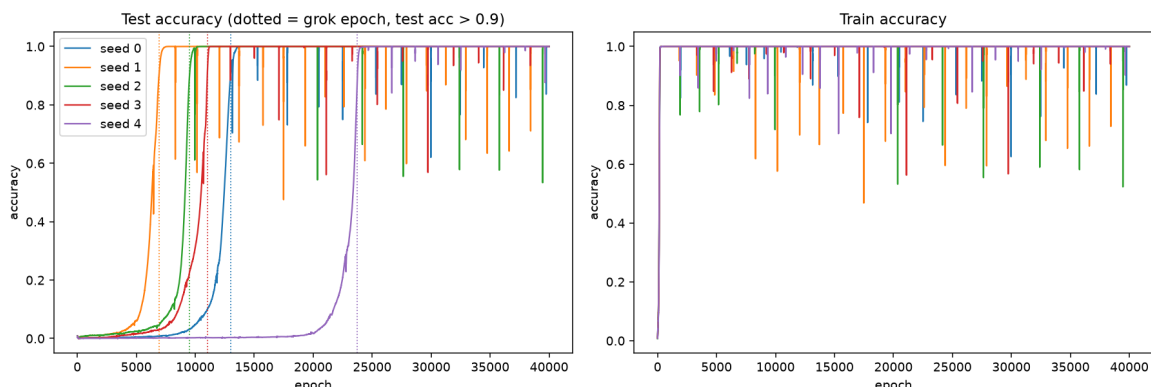


Figure 1. Test and train accuracy for the 5 seeds (dotted lines mark the grok epochs).

3. Pipeline work carried out today

- **Checkpoint schedule:** weights are saved every 50 epochs (801 checkpoints per seed, about 3.2 GB for 5 seeds), so that any 100, 200 or 250 epoch evaluation grid is a subset of what is on disk. Predictors 3 to 5 are computed from these frozen checkpoints without any retraining.

- **Runner path bug fixed:** the repository root and source path in the runner had been broken since the 2026-09-17 folder reorganisation.
- **Magic numbers removed:** the arbitrary constants in the runner now live in the configuration file, each with its source noted.
- **Results and checkpoints committed:** the full 5-seed results and all 4005 checkpoints are in the repository.
- **Shared model:** the model file was not changed today. No earlier predictor code was changed or removed.

4. Predictors and scoring criteria

#	Predictor	Source method	Event read off the signal (as implemented)
1	L2 Norm	Weight-norm moving averages	Moving-average crossover; zero-crossing of the moving average of the slow moving average
2	Dropout	Dropout-variance sweep	Epoch of the variance peak (rate 0.5, 100 samples)
3	Spectral	Canatar et al. 2021, task-model alignment	Minimum of k90; maximum of alignment
4	AGE	Papayan, Han and Donoho 2020, neural collapse NC1	Epoch of minimum NC1
5	HTSR Alpha	Martin and Mahoney 2021, heavy-tailed self-regularisation	Epoch of minimum mean alpha

Two tests, fixed before scoring and applied unchanged to all five predictors. **Test 1:** the event must occur *before* the grok epoch in all 5 seeds (lead = grok epoch minus event epoch, positive means early). **Test 2:** the event epoch must follow the grok epoch across seeds (Spearman rank correlation rho). With n = 5 seeds, rho = 1.0 gives p of about 0.017 and rho = 0.9 gives p of about 0.037; anything below 0.9 is not significant.

5. Results

Predictor and rule	Event epoch, seeds 0 to 4	Before grok	rho
L2: moving-average crossover	113 / 113 / 120 / 130 / 136	5/5	0.70
L2: MA-of-MA zero crossing	1802 / 1767 / 1814 / 1810 / 1801	5/5	0.00
Dropout: variance peak	24200 / 25400 / 16700 / 18800 / 39000	0/5	0.40
Spectral: k90 minimum	29700 / 30700 / 14100 / 38000 / 33900	0/5	0.30
Spectral: alignment maximum	31600 / 37800 / 29300 / 37400 / 32000	0/5	-0.30
AGE: NC1 minimum	31500 / 37800 / 29300 / 37400 / 30100	0/5	-0.40
HTSR: mean alpha minimum	34300 / 16000 / 24100 / 19500 / 39600	0/5	0.90

Table 1. Predictors as implemented, scored on the 5-seed run. Grok epochs are 12987 / 6921 / 9510 / 11019 / 23747.

L2 Norm fires before grok in all 5 seeds, but at almost the same epoch every time (113 to 136 for the crossover, about 1800 for the MA-of-MA rule) while grok itself moves from 6921 to 23747 (rho 0.00 for the MA-of-MA rule). It is detecting an early training transient, not grokking. **Dropout, Spectral, AGE and HTSR** all reach their extreme after grok in every seed: the extreme sits inside the noisy, oscillating post-grok phase or on a flat plateau, so the reported epoch is late and partly arbitrary. HTSR Alpha has a borderline rho of 0.90, but seed 4 drives the correlation and Test 1 already fails.

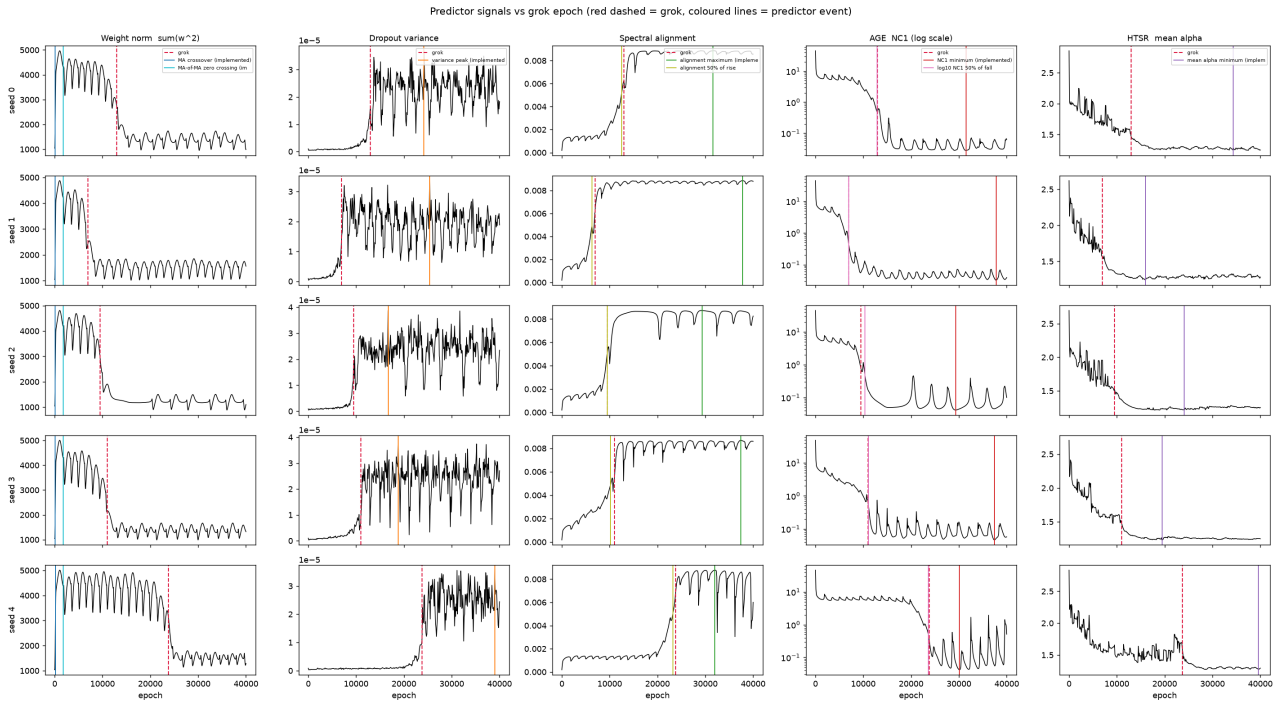


Figure 2. Raw predictor signals for each seed (rows). Red dashed line: grok epoch. Coloured lines: the predictor's event epoch. The signals change around grok; the events fall later.

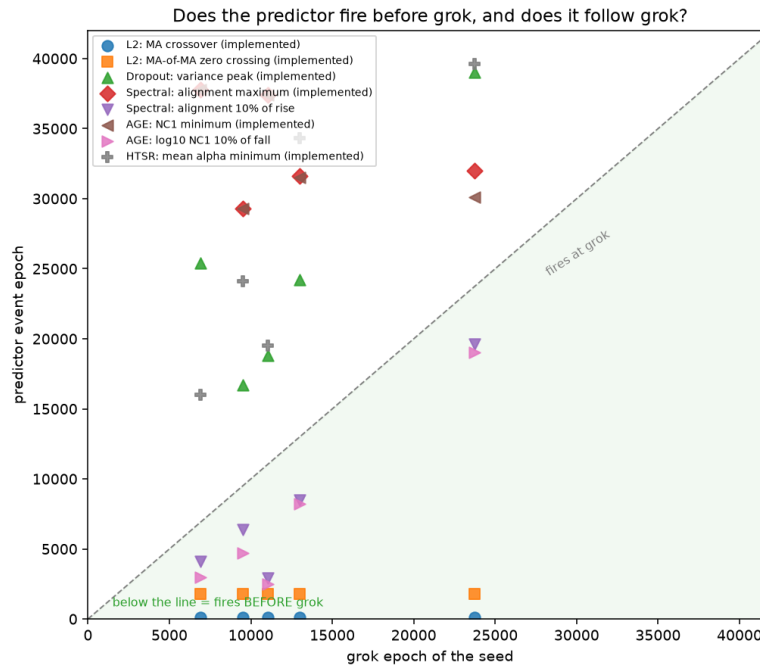


Figure 3. Event epoch against grok epoch. Points below the diagonal would mean the predictor fires before grok.

6. HTSR Alpha (Predictor 5): implementation and caveat

HTSR Alpha follows Martin and Mahoney (2021). For each of the 6 linear weight matrices of the model (query, key, value, mlp_in, mlp_out, output_head) the eigenvalues of $W^T W / N$ are computed, and the exponent alpha of the power-law tail is fitted by **maximum likelihood** with the tail start (xmin) chosen **automatically** by the smallest Kolmogorov-Smirnov distance (Clauset, Shalizi and Newman 2009), using the *powerlaw* package, the same engine WeightWatcher uses. The summary signal is the mean alpha over the 6 layers. No hand-picked quantile or threshold is used.

Seed	Mean alpha first to last	Alpha at grok	Alpha-min epoch (x grok)
0	2.855 to 1.248	1.461	34300 (2.64)
1	2.620 to 1.282	1.569	16000 (2.31)
2	2.696 to 1.253	1.523	24100 (2.53)
3	2.704 to 1.249	1.481	19500 (1.77)
4	2.832 to 1.294	1.581	39600 (1.67)

Caveat. Mean alpha falls from about 2.7 to about 1.25 in every seed, and alpha is already about 1.5 at the grok epoch. However, the fitted alpha is below 2 in 85 to 94 percent of layer-checkpoint fits, whereas the well-trained range in Martin and Mahoney's interpretation is roughly 2 to 6. The layers here are small (at most 128 eigenvalues) and the automatic xmin sometimes keeps a very short tail (as few as 4 eigenvalues in 1 to 4 percent of fits), so the values are noisy and partly outside the range the method was designed for. The numbers are reported as computed; nothing was tuned to move alpha into the 2 to 6 range.

The alpha minimum lies on the flat plateau after grok (mean alpha about 1.25), so which epoch is the minimum is decided by very small wiggles. In seeds 0 and 4 it falls near the end of training.

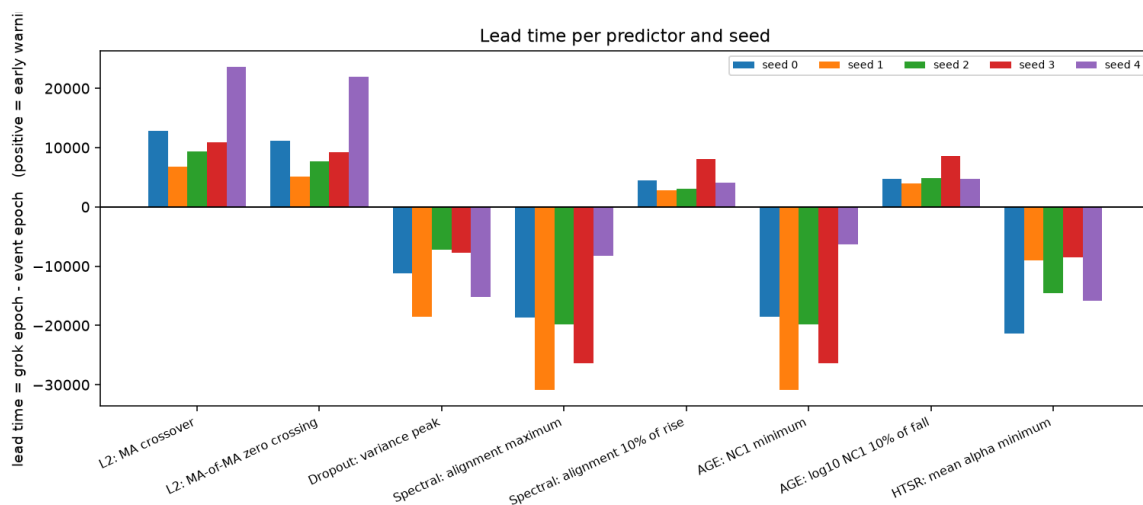


Figure 4. Lead time (grok epoch minus event epoch) per predictor and seed. Negative bars mean the predictor fired after grok.

7. Do the signals themselves carry information?

As a sensitivity check (an **illustrative analysis, not a benchmark result and not a proposed detection rule**), the raw signals were also examined retrospectively: the first epoch at which each signal has covered 10 percent or 50 percent of its total change. The 10 and 50 percent levels are my own arbitrary choice. The 50 percent points of Dropout variance, Spectral alignment and AGE NC1 lie within about 800 epochs of the grok epoch and rank the seeds correctly ($\rho = 1.00$). The 10 percent onsets of Spectral alignment and AGE NC1 occur 2800 to 8500 epochs before grok in all 5 seeds. These quantities use the run's final extreme, so a live monitor could not compute them. The conclusion is limited to this: the signals do change with grokking; the published extreme-value reading is what fails. This analysis was not repeated for HTSR Alpha.

8. Scope decision

Developing an online detection rule (with new thresholds and validation) would turn the thesis from a benchmark of existing predictors into a new-method paper. It has therefore been excluded, and each predictor is evaluated by its published or predefined methodology. The statement "none of the predictors is proven usable" is thus a documented limitation of the present study, not an ignored problem.

9. Limitations

- Only 5 seeds. Only rho of 0.9 or more is significant, and seed 4 (latest grok) dominates several correlations.

- Predictors are evaluated on a 100-epoch checkpoint grid, so timing finer than 100 epochs cannot be resolved.
- A single task and architecture (modular addition, $p = 113$, 1-layer transformer).
- HTSR Alpha values below 2 (see Section 6) mean the method is operating partly outside its designed range on this small model.

10. Next steps and points for discussion

- Continue the evaluation order: **Correlation Traps** (Predictor 6), then Weight-PCA, Higher-MI and Commutator Defect.
- **For discussion:** how to present the HTSR alpha below 2 in the thesis: report as computed with the caveat, or additionally show per-layer alpha, fit quality and tail-size plots?
- **For discussion:** whether the retrospective sensitivity analysis of Section 7 should appear in the thesis at all, given the scope decision.